Cross-Scene Open-Set Hyperspectral Image Classification via Joint Distribution Matching and Unknown-Class Uncertainty Suppression

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***Abstract*—Cross-scene hyperspectral image (HSI) classification plays a vital role in a variety of applications, including agricultural monitoring, military reconnaissance, and urban planning. Due to the nonlinear nature of land-cover characteristics, the diversity of scenes, and the distribution shifts caused by variations in acquisition conditions or sensors, deep HSI open-set domain adaptation (OSDA) has emerged as a key approach to tackling cross-scene HSI classification. Existing methods assume identical joint distributions between training and test data. However, this rigorous assumption renders them ineffective in real-world scenes when encountering varied sensors, changed imaging conditions, different spatial resolution in HSI imaging, e.g., water vapor bands at 1.36–1.39 μm. To address this, we propose a more generalizable deep HSI OSDA method, JDMUS, based on joint distribution matching (JDM) and unknown-class uncertainty suppression (US), for cross-scene HSI classification. Specifically, we employ χ^2-divergence to characterize the joint distribution discrepancy and propose a least-squares χ^2-divergence estimator without explicitly modeling the two joint distributions, yielding a quadratic problem with an analytical solution. Furthermore, we prove the methodology that the upper bound of the target error for a neural network classification model is determined by the average source error, χ^2-divergence, and open-set risk difference. Based on this methodology, we design JDM and unknown-class US mechanisms and thus construct the JDMUS algorithm with theoretical guarantees, in which a weighted generative adversarial network with dynamic thresholding (WGDT) is used for feature extraction. Experiments on cross-scene HSI datasets demonstrate that our JDMUS outperforms existing state-of-the-art approaches.**

***Index Terms*—Cross-scene hyperspectral image (HSI) classification, joint distribution matching (JDM), open-set domain adaptation (OSDA), transfer learning, unknown-class uncertainty suppression (US).**

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I. Introduction

HYPERSPECTRAL image (HSI) classification exhibits extensive application value across diverse fields, particularly in environmental and ecological monitoring [1], [2], precision agriculture [3], [4], and geological exploration [5], [6], owing to its ability to capture fine-grained spectral signatures. However, constructing a training set and classification model for every new scene is impractical, since pixel-level HSI annotation, which relies heavily on expert knowledge, is both expensive and time-consuming. Fortunately, the same crop tends to present similar spectral characteristics in different scenes. For example, an agricultural classification model can be transferred from one farmland scene to another. This has motivated increasing research interest in cross-scene HSI classification in recent years.

Benefiting from its ability to handle the nonlinear nature of land-cover characteristics and the distribution shifts caused by variations in acquisition conditions or sensors, deep HSI domain adaptation (DA) technique [7], [8] naturally plays an important role in cross-scene HSI classification. Traditional deep HSI DA methods assume that the source and target domains share identical label spaces, known as closed-set DA (CSDA) [9], [10], [11], [12]. Fang et al. [9] proposed a confident learning-based domain adaptation framework (CLDA) that effectively transfers discriminative spectral–spatial features across domains. This approach enhances adaptation reliability by explicitly reducing the impact of low-confidence samples. Li et al. [10] introduced a supervised contrastive learning-based unsupervised DA (SCLUDA) method, which leverages contrastive objectives to

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improve representation robustness in HSI classification. This method can demonstrate strong generalization ability but often requires large batch sizes to ensure sufficient positive–negative sample pairs, which may increase computational cost. Huang et al. [11] developed an adversarial discriminative DA model that aligns source and target distributions through adversarial learning while preserving class separability. In addition, Huang et al. [12] designed a two-branch attention adversarial network (TAADA) that integrates spectral and spatial attention mechanisms into adversarial DA, thereby enhancing feature discriminability and domain invariance.

These deep CSDA approaches have achieved remarkable performance improvements in mitigating domain discrepancies. However, this identical label space assumption of source and target domains often fails in real-world scenarios. Domain shifts caused by atmospheric variations [13] and sensor-specific noise patterns [14] significantly degrade model performance when applied to new target domains, where the target domain may contain unseen classes in the source domain. The aerosol scattering effect, like Rayleigh and Mie scattering mechanisms, induces systematic distortions in surface reflectance, particularly within short-wavelength regions [15]. When atmospheric visibility in the target domain is 30% lower than the source domain, spectral features of urban construction materials exhibit significant attenuation of characteristic absorption peaks [16], which generates pseudo-unknown-class signatures. This more realistic setting, called HSI open-set DA (OSDA), poses significant challenges as direct alignment of domains containing unknown classes can lead to negative transfer phenomena [17].

Current deep HSI OSDA approaches can be categorized into three mainstream categories. Bi et al. [18] proposed a weighted generative adversarial network with dynamic thresholding (WGDT) framework for HSI OSDA. By integrating class-anchor (CA) reliability weights with a learnable threshold, this method effectively mitigates negative transfer and improves rejection of unknown classes. Tang et al. [19] developed a multiclassifier adversarial network for HSI OSDA, in which multiple classifiers with diversity constraints enhance boundary calibration and increase robustness to unknown classes. Li et al. [20] introduced a pixel-aware weighting and decoupled alignment framework for HSI OSDA. By assigning adaptive weights to target samples and decoupling marginal from conditional alignment, this method reduces false alignment of unknown pixels. Nirmal et al. [21] proposed a generative adversarial network-based OSDA method for HSI classification, focusing on adversarial domain alignment and open-set rejection under significant spectral distribution shifts. Gao et al. [22] proposed the threshold-domain adversarial network, originally for generic OSDA, but widely adopted in HSI classification. It progressively selects transferable target samples based on adaptive thresholds, alleviating negative transfer. Saito et al. [23] proposed open-set backpropagation (OSBP), which extends adversarial DA with an additional classifier to explicitly separate known and unknown classes, thereby providing one of the earliest baselines for OSDA. Tan et al. [24] introduced collaborative distribution alignment, where multiple classifiers collaboratively predict and align target samples, improving robustness to unknown categories while mitigating negative transfer.

Despite the advances of these deep OSDA approaches in tackling cross-scene HSI classification, there still exists a critical limitation. They assume identical joint distributions between training and test data. However, this rigorous assumption is hard to hold in real-world scenarios due to varied sensors, changed imaging conditions, and different spatial resolutions in HSI imaging. Therefore, from the perspective of theoretical investigation, according to the realistic premise that the joint distributions between source and target domains are not consistent, studying a reliable joint distribution alignment method for cross-scene open-set HSI classification is challenging but valuable.

In this study, we propose a novel OSDA method, JDMUS, based on joint distribution matching (JDM) and unknown-class uncertainty suppression (US) for cross-scene open-set HSI classification. Specifically, we systematically investigate a new and tighter generalization error bound for target DA based on joint distribution alignment in open-set scenarios, which exhibits that the upper bound of the target error for a neural network classification model is determined by the average source error, χ^2-divergence, and open-set risk difference. Under this upper bound guarantee, we present JDM and unknown-class US mechanisms, utilize weighted generative adversarial network with dynamic thresholding to extract features, and thus construct the deep HSI OSDA algorithm JDMUS.

The main contributions are summarized as follows.

1) For cross-scene HSI classification, we propose a novel deep HSI OSDA method, JDMUS, with strong generalization capability and high interpretability. It theoretically handles the vital JDM and unknown-class US problems.
2) To ensure superior interpretability, we establish a new theoretical error bound for OSDA. The theory demonstrates that the upper bound of the target error in neural networks is determined by three factors: the source domain risk of the network, the χ^2-divergence between the joint distributions of the source domain and the known classes in the target domain, and the open-set difference.
3) To further achieve strong generalization capability, we construct the JDMUS algorithm under this new OSDA error bound. From the risk of empirical distribution discrepancy, we derive the JDM mechanism to enable reliable knowledge transfer at both the semantic and feature levels, which reduces the incidence of negative transfer and improves the robustness of models when adapting to unseen environments. Meanwhile, from the risk of open-set difference estimation, we derive an unknown-class US mechanism to avoid the classification performance fluctuation caused by blindly recognizing target data as unknown classes.
4) Experiments are conducted on four different cross-scene HSI datasets, showing that JDMUS can effectively identify unknown samples while maintaining excellent

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HSI classification accuracy on known classes, achieving SOTA performance compared to existing methods.

II. Related Work

A. Closed-Set DA

Existing CSDA methods are mainly divided into three categories. The first category is based on the covariate shift assumption. Sugiyama et al. [25] proposed the Kullback–Leibler (KL) importance estimation procedure, which estimates the weight function by minimizing the KL divergence between the target distribution and the weighted source distribution. Bickel et al. [26] treated source and target data as two classes and learned sample weights via logistic regression, enabling the construction of weighted prediction models. Building on these ideas, Cao et al. [27] introduced selective transfer machine and distribution matching machine frameworks, while Chen et al. [28] designed two subspace adaptation methods under the same covariate shift assumption. These methods assume identical conditional probabilities across domains and are effective for marginal distribution alignment. The second category focuses on domain-invariant representations. Pan et al. [29] proposed transfer component analysis, which aligns marginal distributions by minimizing maximum mean discrepancy. Baktashmotlagh et al. [30] developed distribution matching embedding, which learns feature mappings by minimizing maximum mean discrepancy or Hellinger distances. These methods focus on extracting domain-invariant representations that can generalize across different domains. The third category directly aligns joint probability distributions. Courty et al. [31] introduced joint distribution optimal transportation, the first approach to align joint distributions via optimal transportation while simultaneously minimizing target error bounds. Damodaran et al. [32] extended this idea with DeepJDOT, which leverages CNNs and progressive coupling strategies to improve scalability. Chen et al. [33] further generalized joint distribution alignment under f-divergence, deriving a closed-form solution via quadratic convex optimization. Specific to HSI cross-scene CSDA, Zhao et al. [34] investigated an unsupervised domain adaptation framework for cross-scene HSI classification, which performs multi-level feature alignment together with decision boundary adaptation to effectively mitigate spectral distribution shifts across scenes. Ye et al. [35] proposed a feature-level cross-scene domain adaptation approach for HSI classiffcation based on dictionary learning, where a common dictionary is learned to project source and target samples into a domain-invariant representation space. Our method instead considers the HSI OSDA problem, which is more common in real scenarios due to diverse imaging interferences.

B. Open-Set DA

Recent advancements in OSDA have introduced diverse methodologies to address domain shift while identifying unknown-class samples. Saito et al. [23] proposed open-set backpropagation, which employs adversarial learning with a binary classifier to reject unknown samples. Shermin et al. [36] improved robustness by introducing domain adversarial learning with multiple classiffers (DAMC). Liu et al. [37] presented a separate-to-adapt (STA) methodology, which aligns known-class prototypes while separating unknowns through angular margin constraints in embedding space. Jang et al. [17] developed unknown-aware domain adversarial learning (UADAL), leveraging Monte Carlo dropout to estimate predictive uncertainty and dynamically filter out unknown categories. Chang et al. [38] proposed a mutual-to-separate (MTS) framework for open-set domain adaptation, which first aligns representations through mutual learning among multiple classiffers and then explicitly separates unknown target samples from known classes based on prediction inconsistency. Zhao and Shen [39] proposed a dual adversarial network for cross-domain open-set fault diagnosis (DACD), which employs adversarial learning to achieve domain-invariant feature alignment while handling unknown target faults through an open-set discrimination mechanism. Li et al. [40] proposed an adjustment-and-alignment framework (ANNA) for unbiased open-set domain adaptation, in which a dedicated adjustment mechanism is introduced to mitigate the bias caused by unknown target samples before performing cross-domain alignment. Ru et al. [41] proposed open-set moving-threshold estimation and gradual alignment (OMEGA), which dynamically estimates decision thresholds to identify reliable target samples, and progressively performs domain alignment under imbalanced open set settings. Collectively, these methods target the core challenges of OSDA, including reliable unknown-class detection, domain-invariant representation learning, and the design of adaptive decision boundaries. Nejjar et al. [42] proposed recall and refinement for domain adaptation (RRDA), a source-free OSDA framework that recalls class-discriminative representations from a pretrained source model and refines decision boundaries on the target domain without access to labeled source data. Silva et al. [43] proposed an unknown exploration framework that explicitly identifies unknown target samples and incorporates them into the adaptation process to improve decision boundary separation in open-set domain adaptation. Huang and Ren [44] proposed balanced correlation learning, which incorporates label-prior-aware correlation objectives based on the Hilbert–Schmidt independence criterion to mitigate open-set bias and promote more reliable alignment in open-set domain adaptation. Chaudhari et al. [45] addressed distribution changes even for known classes, jointly learning known classification and unknown detection with formal robustness guarantees. However, they do not consider the issue of aligning the joint distribution of source and target domains, which limits its generalization performance in real HSI scenarios. To mitigate this, our method establishes joint distribution alignment between the source and target domains based on χ^2-divergence, and derives a new and tighter generalization error bound control for HSI OSDA.

C. Weighted Generative Adversarial Networks With Dynamic Thresholding

The OSDA framework WGDT [18] for HSI classification is the most closely related work to this study, and it facilitates

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the feature extraction. Specifically, WGDT consists of four coupled components.

1) *Dual-Channel Feature Extractor ϕ:* A dual-channel residual network is employed to capture both spectral and spatial representations from hyperspectral cubes. The extracted features provide a compact and discriminative embedding space for subsequent adaptation.
2) *CA Weighting Strategy:* For each known class in the source domain, a class anchor is constructed in the learned metric space. By measuring the distance between a target sample and these anchors, a reliability score is derived. This score is then normalized through a weight calculation module to produce instance-level reliability weights, which quantify the likelihood that a target sample belongs to a known class. These weights guide domain alignment and threshold learning, ensuring that samples similar to unknown classes exert less influence.
3) *Instance-Level Weighted Adversarial Learning:* Unlike conventional domain adversarial learning, which treats all target samples equally, WGDT integrates the CA-derived weights into the adversarial loss. Samples with higher reliability contribute more to alignment.
4) *Dynamic-Threshold Classifier g:* The classifier consists of two linked components. First, a minimum rejection score calculation module computes the rejection score $\beta_{\min}$ for each target sample, reflecting its minimum distance to all class anchors. Second, a learnable threshold parameter R is optimized during training to approximate the decision boundary between known and unknown samples. Specifically, a target sample is classified as unknown if $\beta_{\min} > R$, and assigned to the nearest known class otherwise. This dynamic thresholding mechanism enables the model to adaptively separate known from unknown categories across diverse domains.

Through the combination of the above components, WGDT reduces distributional shift and maintains robustness against unseen classes in the target domain to some extent. Different from WGDT, our method introduces key distinctions by matching the joint probability distribution of known-class samples across domains and mitigating the open-set difference. This critical enhancement allows our model to further alleviate the domain shift and address the problem of high uncertainty in classifying target unknown samples near decision boundaries.

III. Methodology

In this section, we establish a new theoretical error bound for open-set generalization to ensure the interpretability and superiority of methodology.

A. Problem Definition

Let $\mathcal{X} \subset \mathbb{R}^D$ be the feature space, and $\mathcal{Y}^s$, $\mathcal{Y}^u$, and $\mathcal{Y}^t$ be the label spaces, satisfying $\mathcal{Y}^s, \mathcal{Y}^u \subset \mathcal{Y}^t$, $\mathcal{Y}^s \cap \mathcal{Y}^u = \emptyset$, and $\mathcal{Y}^s \cup \mathcal{Y}^u = \mathcal{Y}^t$. $\mathcal{X}$ is shared in the source and target domains. $\mathcal{Y}^s = \{y_c^s\}_{c=1}^{C}$ and $\mathcal{Y}^t = \{y_c^t\}_{c=1}^{C+1}$ represent the source and target domains' label spaces, respectively. $\mathcal{Y}^u = \{y_{C+1}^t\} = \{y^u\}$ represents the unknown class in target domain. y_c^s is the cth class in source domain, y_c^t is the cth class in the target domain, and y^u is the unknown class in target domain.

Let $X^s, X^t \in \mathcal{X}$ be the random variables with labels $Y^s \in \mathcal{Y}^s$ and $Y^t \in \mathcal{Y}^t$. The joint probability distribution $P(X^s, Y^s)$ is defined on the product space $\mathcal{X} \times \mathcal{Y}^s$ for the source domain, while $P(X^t, Y^t)$ is defined on $\mathcal{X} \times \mathcal{Y}^t$ for the target domain. For convenience, we denote $P(X^s, Y^s)$ as $P_{X^sY^s}$, $P(X^t, Y^t)$ as $P_{X^tY^t}$, the marginal distribution $P(X^s)$ as P_{X^s}, and the marginal distribution $P(X^t)$ as P_{X^t}.

Definition 1 (Pen-Set Domain Adaption): Suppose $P_{X^sY^s} \neq P_{X^tY^t}$. Given a labeled dataset $D^s = \{(x_i^s, y_i^s)\}_{i=1}^m$ sampled from the source joint distribution $P_{X^sY^s}$, and an unlabeled dataset $D^t = \{(x_i^t)\}_{i=1}^m$ sampled from the target marginal distribution P_{X^t}, the unsupervised OSDA intends to select an optimal model h

$$\left\{\mathcal{X} \to \mathbb{R}^{|\mathcal{Y}|} \to \mathbb{R}^{c_s+1}, x \to \left[h_1(x), \ldots, h_{c_s+1}(x)\right]^T\right\}$$

such that the generalization error on the target domain is minimized, where ℓ is the symmetric loss function

$$E_{(x,y)\sim P_{X^tY^t}(x,y)}\left[\ell\left(h(x), y\right)\right] = \int_{\mathcal{X}\times\mathcal{Y}^t} \ell\left(h(x), y\right) P_{X^tY^t}(x,y)\, dxdy. \tag{1}$$

Definition 2 (Risks and Partial Risks): In statistical theory, risk and partial risk are two core concepts for evaluating model performance, reflecting the generalization capability of models in data prediction. According to Definition 1, our task is a multiclass classification problem aiming to find the optimal classification model h in hypothesis space $\mathcal{H}$, where h_c represents the confidence score of a sample belonging to class$c$. For a model $h \in \mathcal{H}$ constructed under the joint distribution of the source domain and the target domain, we give the following risk definitions:

$$R^s(h) = \int_{\mathcal{X}\times\mathcal{Y}^s} \ell\left(h(x), y\right)\, P_{X^sY^s}(x,y)\, dx\ dy \tag{2}$$

$$R^t(h) = \int_{\mathcal{X}\times\mathcal{Y}^t} \ell\left(h(x), y\right)\, P_{X^tY^t}(x,y)\, dx\ dy. \tag{3}$$

The partial risks for samples from the source and target domains being labeled as the unknown class c are, respectively, defined as $R_u^s(h)$ and $R_u^t(h)$

$$R_u^s(h) = \int_{\mathcal{X}} \ell\left(h(x), y^u\right)\, P_{X^s}(x)\, dx \tag{4}$$

$$R_u^t(h) = \int_{\mathcal{X}} \ell\left(h(x), y^u\right)\, P_{X^t}(x)\, dx. \tag{5}$$

B. Error Bound

To assess the dissimilarity between $P_{X^sY^s}$ and $P_{X^tY^t}$, we adopt χ^2-divergence

$$D_{CS}\left(P_{X^sY^s} \,\|\, P_{X^tY^t}\right) = \int P_{X^sY^s}(x,y)\left(\frac{P_{X^sY^s}(x,y)}{P_{X^tY^t}(x,y)}\right) dxdy - 1, \tag{6}$$

where the χ^2-divergence is an f-divergence measuring the discrepancy between two probability distributions. In this study, we compute the divergence between joint probability distributions of shared classes across source and target

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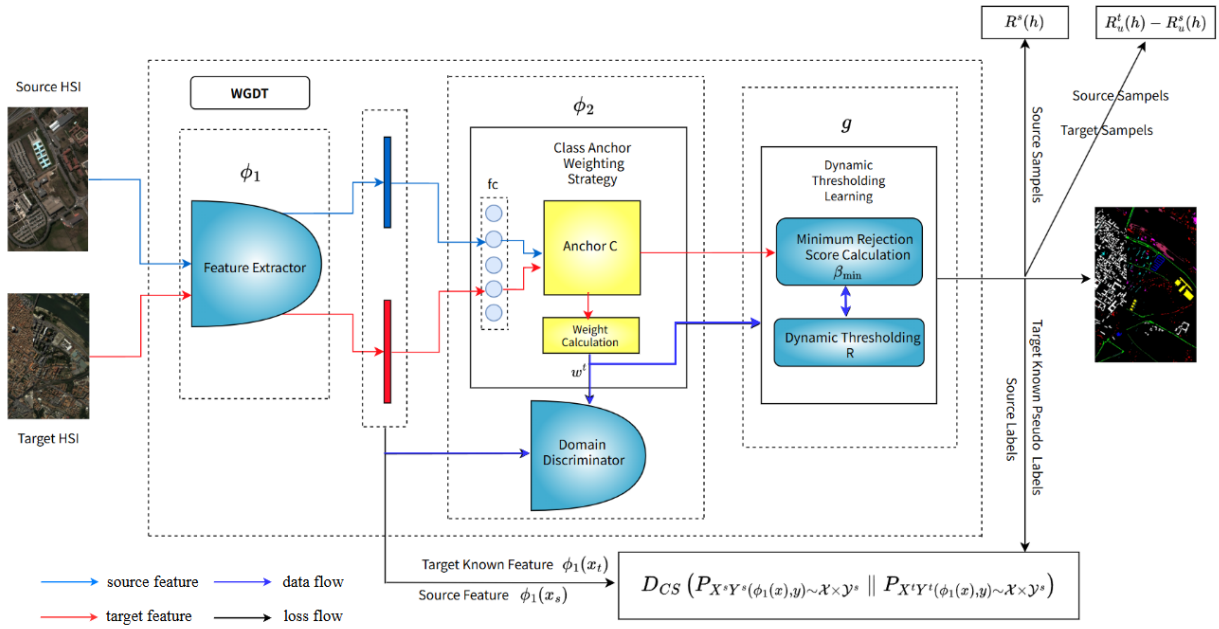


Fig. 1. Overview of JDMUS. Motivated by WGDT [18], JDMUS also consists of four fundamental operators: dual-channel feature extractor, CA weighting strategy, instance-level weighted adversarial learning, and dynamic-threshold classifier. Different from WGDT, under the guarantee of our proposed tight generalization error bound, JDMUS is along with three gain modules: RS control, joint distribution discrepancy control, and open-set difference control.

domains using χ^2-divergence, which is given by the following formula:

$$D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \,\|\, P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right) = \int_{\mathcal{X}\times\mathcal{Y}^s} P_{X^sY^s}(x,y)\left(\frac{P_{X^sY^s}(x,y)}{P_{X^tY^t}(x,y)}\right)dxdy - 1 \tag{7}$$

where $P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s}$ and $P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}$, respectively, denote the joint probability distributions of the source domain and the shared classes in the target domain, both defined on the product space $\mathcal{X}\times\mathcal{Y}^s$.

Theorem 1 (Error Bound): Assume the loss function satisfies $\ell(h(x),y)\le M$ for $y\in\mathcal{Y}^s$ with $M>0$, and $\ell(h(x),y^u)\le N$ for $y^u\in\mathcal{Y}^u$ with $N>0$. For any hypothesis $h:\mathcal{X}\to\mathbb{R}^{|\mathcal{Y}|}\to\mathbb{R}^{c_s+1}$ that maps x to $[h_1(x),\ldots,h_{c_s}(x),h_{c_s+1}(x)]^\top$, the generalization error in the target domain is bounded by

$$R^t(h) \le \overbrace{R^s(h)}^{\text{Source Risk}} + \overbrace{R_u^t(h)-R_u^s(h)}^{\text{Open-Set Difference}} + (M+N)\underbrace{\sqrt{2D_{\text{CS}}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \,\|\, P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)}}_{\text{Distribution Discrepancy}}. \tag{8}$$

Proof: Please see Appendix.

Remark 1: From Theorem 1, to train an OSDA model achieving low target-domain error, the algorithm should minimize the following three terms.

1) *Source Risk*: Reducing the model's classification risk on the source domain to enhance knowledge transfer to the target domain.
2) *Open-Set Difference*: Minimizing the risk discrepancy between the target and source domains for samples labeled as "unknown", i.e., larger positive term indicates more target samples being classified as unknown, while a negative term prevents source samples from being misclassified as unknown.
3) *Distribution Discrepancy*: Aligning the joint probability distributions of shared classes across domains to improve classification accuracy for known classes in target domain.

IV. Algorithm

In this section, based on the new error bound methodology presented above, we propose a novel deep HSI OSDA algorithm, JDMUS, for cross-scene HSI classification. Specifically, our classification framework employs a neural network model $h:\mathcal{X}\to\mathcal{Y}$, which consists of two fundamental components: 1) the feature extractor $\phi:\mathcal{X}\to\mathcal{F}$ that projects input samples into an intermediate feature space and 2) the probabilistic classifier $g:\mathcal{F}\to\mathcal{Y}$ responsible for final classification, with the complete model architecture formally defined by the composition $h=g\circ\phi$ for all $x\in\mathcal{X}$.

A. Proposed Framework

An overview of our algorithm is illustrated in Fig. 1. The neural network model h within the WGDT framework [18] consists of four modules: the feature extractor $\phi=\phi_2\circ\phi_1$, where ϕ_1 is a spectral–spatial feature extractor and ϕ_2 integrates a CA strategy with instance-level weighted adversarial learning, and a dynamic-threshold classifier g. Source- and target-domain data, x^s and x^t, are first fed into ϕ_1 to obtain features $\phi_1(x^s)$ and $\phi_1(x^t)$, which are then processed by the CA operator and domain adversarial learning ϕ_2 to generate credibility weights indicating known-class confidence. However, as this feature extraction may not ensure JDM of known

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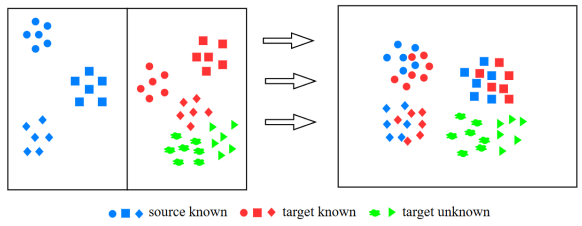


Fig. 2. Illustration of divergence computation restricted to the known-class subspace.

classes between domains, which can lead to negative transfer, a joint distribution discrepancy control is introduced during spectral feature extraction. We introduce an unknown-class US module to reduce the misclassification of source known-class samples as unknown and improve the detection of target-domain unknown classes. This is achieved by applying an open-set difference control, $\hat{R}_u^t(h) - \gamma \hat{R}_s(h)$, to enhance the generalization ability of the classifier g. The empirical risk $\hat{R}_s(h)$ for g is computed on the source samples. In addition, pseudo-labels p are generated to quantify the distribution discrepancy of known classes across domains. To mitigate negative transfer, we restrict divergence computation to the *known-class subspace*, as illustrated in Fig. 2. Only feature representations corresponding to reliable source classes participate in alignment, while target features predicted as potential unknowns are excluded and optimized through the open-set risk term. This decoupling ensures that alignment gradients are propagated only within the shared subspace, avoiding inadvertent attraction of unknown samples toward known-class anchors. Consequently, the DA process remains stable and discriminative across both known and unknown regions. Thus, the optimization problem of JDMUS can be formulated as follows:

$$\min_{\phi,g} \hat{R}^s(g \circ \phi) + \hat{R}_u^t(g \circ \phi) - \gamma \hat{R}_u^s(g \circ \phi) + \alpha \sqrt{D_{CS}\left(P_{X^sY^s(\phi_1(x),y)\sim \mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t(\phi_1(x),y)\sim \mathcal{X}\times\mathcal{Y}^s}\right)} \quad (9)$$

where γ and α are the tradeoff parameters, and $\hat{R}^s(h)$, $\hat{R}_u^t(h)$, and $\hat{R}_u^s(h)$ represent the empirical estimation.

The empirical risks of $R^s(h)$ can be estimated by $\hat{R}^s(h)$

$$\hat{R}^s(h) = \frac{1}{m_s} \sum_{i=1}^{m_s} \ell\left(g\left(\phi\left(x_i^s\right)\right), y_i^s\right). \quad (10)$$

Building upon the WGDT framework, we develop an enhanced empirical risk estimation approach for OSDA in HSI classification. The source-domain risk estimation follows the baseline implementation, calculating the average loss over all labeled source samples.

B. Joint Distribution Matching

In DA, aligning the source and target domains solely at the feature level often proves insufficient, as semantic consistency across domains must also be preserved to ensure reliable knowledge transfer. JDM has, therefore, become a key strategy, as it enforces alignment between domains in a way that captures both structural characteristics of the data and task-related information. This alignment reduces the risk of negative transfer and improves the robustness of models when adapting to unseen environments.

Pseudo-labels [46] are widely utilized in DA. During the joint distribution alignment phase, the label information from the target domain is required, where the WGDT dynamic thresholding module generates predictive outputs for target domain samples as pseudolabels. In cross-scene open-set adaptation, the early stage often suffers from instability due to unreliable pseudolabels from the target domain. To mitigate this, we adopt a two-stage schedule distinguishing early and late learning dynamics. During the warm-up stage, the model performs source-domain fine-tuning; if no reliable known-class samples are identified, the alignment loss is skipped to avoid noisy gradients. A strict confidence threshold τ initially filters target samples, and as the model becomes calibrated, τ is linearly relaxed to gradually include moderate-confidence samples. This ramp-up stage enables a smooth transition from cautious alignment to full optimization, achieving both stability and completeness in adaptation. The empirical form of the divergence can be written as

$$D_{CS}\left(P_{X^sY^s} \parallel P_{X^tY^t}\right) = \frac{1}{m_s} \sum_{i=1}^{m_s} \frac{P_{X^sY^s}\left(\phi_1\left(x_i^s\right), y_i^s\right)}{P_{X^tY^t}\left(\phi_1\left(x_i^s\right), y_i^s\right)} - 1. \quad (11)$$

As the density ratio function $(P_{X^sY^s}(\phi_1(x), y))/(P_{X^tY^t}(\phi_1(x), y))$ is unknown, we adopt the least-squares fitting method proposed in [47] to directly estimate it. We model the ratio function as a linear combination of basis functions

$$r(\phi_1(x), y; \rho) = \sum_{j=1}^{m_s+m_{tk}} \rho_j \Phi_j(\phi_1(x), y), \quad (12)$$

where $\rho = (\rho_1, \ldots, \rho_{m_s+m_{tk}})^\top$ are the unknown parameters to be estimated, m_s is the number of source-domain samples, m_{tk} is the number of target domain samples with known classes, and $\Phi_j(\phi_1(x), y)$ is the jth basis function.

The parameters ρ can be estimated by minimizing the following objective function, where Const represents constant terms independent of ρ:

$$\begin{aligned} J(\rho) &= \frac{1}{2} \int \left(r(\phi_1(x), y; \rho) - \frac{P_{X^sY^s}(\phi_1(x), y)}{P_{X^tY^t}(\phi_1(x), y)} \right)^2 P_{X^tY^t}(\phi_1(x), y)\, dxdy \\ &= \frac{1}{2} \int r^2(\phi_1(x), y; \rho) P_{X^tY^t}(\phi_1(x), y)\, dxdy \\ &\quad - \int r(\phi_1(x), y; \rho) P_{X^sY^s}(\phi_1(x), y)\, dxdy + \text{Const}. \end{aligned} \quad (13)$$

Since the right-hand side of (13) is a mathematical expectation, we can accordingly approximate the function $J(\rho)$ using sample means

$$\widehat{J}(\rho) = \frac{1}{2} \sum_{i=1}^{m_{tk}} r^2\left(x_i^t, y_i^t; \rho\right) - \sum_{i=1}^{m_s} r\left(x_i^s, y_i^s; \rho\right) + \text{Const}', \quad (14)$$

where the optimal parameters ρ can be obtained by solving the following ridge-regularized optimization problem. The penalty

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term $\lambda/2\rho^\top\rho$ serves as a regularization component, where $\lambda \geq 0$ represents the regularization parameter

$$\hat{\rho} = \underset{\rho\in\mathbb{R}^{m_s+m_{tk}}}{\arg\min}\left[\widehat{J}(\rho) + \frac{\lambda}{2}\rho^\top\rho\right] = \underset{\rho\in\mathbb{R}^{m_s+m_{tk}}}{\arg\min}\left[\frac{1}{2}\rho^\top H\rho - h^\top\rho + \frac{\lambda}{2}\rho^\top\rho\right]. \tag{15}$$

Here, H is a $(m_s + m_{tk}) \times (m_s + m_{tk})$ matrix with its (j,k)th element given by

$$H_{jk} = \int \Phi_j\left(\phi_1(x), y\right)\Phi_k\left(\phi_1(x), y\right)P_{X^tY^t}\left(\phi_1(x), y\right)dxdy = \frac{1}{m_{tk}}\sum_{i=1}^{m_{tk}}\Phi_j\left(\phi_1\left(x_i^t\right), y_i^t\right)\Phi_k\left(\phi_1\left(x_i^t\right), y_i^t\right) \tag{16}$$

and h is a $(m_s + m_{tk})$-dimensional vector with its jth element

$$h_j = \int \Phi_j\left(\phi_1(x), y\right)P_{X^sY^s}\left(\phi_1(x), y\right)dxdy = \frac{1}{m_s}\sum_{i=1}^{m_s}\Phi_j\left(\phi_1\left(x_i^s\right), y_i^s\right). \tag{17}$$

Thus, (15) has the closed-form solution

$$\hat{\rho} = \left(\hat{H} + \lambda I\right)^{-1}\hat{h} \tag{18}$$

where I is an identity matrix. Consequently, the joint distribution ratio function $(P_{X^sY^s}(\phi_1(x), y))/(P_{X^tY^t}(\phi_1(x), y))$ can be estimated as

$$r\left(\phi_1(x), y; \hat{\rho}\right) = \sum_{i=1}^{m_s+m_{tk}}\hat{\rho}_i\Phi_i\left(\phi_1(x), y\right). \tag{19}$$

To prevent numerical instability during computation when substituting the ratio function in (11) with $r(\phi_1(x), y; \hat{\rho})$, we impose a nonnegativity constraint: the operation $\max[r(\phi_1(x), y; \hat{\rho}), \epsilon]$, where ϵ is a small positive constant. The basis functions $\Phi_j(\phi_1(x), y)$ are designed as $k(\phi_1(x), \phi_1(x_j))l(y, y_j)$, where $k(\phi_1(x), \phi_1(x_j))$ is a kernel function for the features, $l(y, y_j)$ is a kernel function for the labels, and $(\phi_1(x_j), y_j)$ is the jth element of $D^s \cup D^{tk}$. For the feature, we utilize the Gaussian kernel with bandwidth $\sigma_x > 0$ set to the median of pairwise distances

$$k\left(\phi_1(x), \phi_1\left(x_j\right)\right) = \exp\left(-\frac{\left\|\phi_1(x) - \phi_1\left(x_j\right)\right\|^2}{\sigma_x}\right). \tag{20}$$

For the label, we employ the delta kernel

$$l\left(y, y_j\right) = \begin{cases}1, & y = y_j\\ 0, & \text{otherwise.}\end{cases} \tag{21}$$

Based on the above analysis, for superior JDM to mitigate negative transfer, the empirical divergence between the joint distributions of the source and target domains is expressed as the third term in (8)

$$D_{CS}\left(P_{X^sY^s} \parallel P_{X^tY^t}\right) = \frac{1}{m_s}\sum_{i=1}^{m_s}\max\left\{r\left(\phi_1\left(x_i^s\right), y_i^s; \left(\hat{H} + \lambda I\right)^{-1}\hat{h}\right), \epsilon\right\} - 1. \tag{22}$$

Algorithm 1 Source Pretraining

Input: Labeled source dataset D^s.
Parameter: batch size b; maximum epochs K_s.
Output: Pretrained feature extractor ϕ and classifier g.

```
Initialize network parameters of φ and g.
for epoch = 1 to K_s do
    Sample a minibatch {(x_s, y_s)} from D^s.
    Compute source predictions g(φ(x_s)).
    Compute supervised loss L_s = ℓ(g(φ(x_s)), y_s).
    Update φ and g via gradient descent with learning rate η.
end for
return pretrained φ, g.
```

C. Unknown-Class US

We introduce an unknown-class US module, aiming to mitigate the misclassification of source known-class samples as unknown and enhance the detection of unknown classes in the target domain by controlling the open-set difference.

We expect to explicitly control the open-set difference, i.e., minimizing the risk difference between target and source samples that are identified as "unknown." A large positive discrepancy term $R_u^t(h)$ indicates that more target-domain samples are being categorized as unknown, whereas a negative discrepancy $R_u^s(h)$ prevents source-domain samples from being misclassified as unknown.

The empirical risks of $R_u^s(h)$ and $R_u^t(h)$ can be estimated by $\hat{R}_u^s(h)$ and $\hat{R}_u^t(h)$, respectively,

$$\hat{R}_u^s(h) = \frac{1}{m_s}\sum_{i=1}^{m_s}\ell\left(g\left(\phi\left(x_i^s\right)\right), y_i^u\right) \tag{23}$$

$$\hat{R}_u^t(h) = \frac{1}{m_t}\sum_{i=1}^{m_t}\ell\left(g\left(\phi\left(x_i^t\right)\right), y_i^u\right). \tag{24}$$

Bi et al. [18], to calculate $\ell(g(\phi(x_i^s)), y_i^u)$, classified the samples satisfying $\beta_{\min}(g(\phi(x_i^s))) > \tau$ as the unknown class, where $\beta_{\min} = \min(\beta)$, $\beta = d \circ (1 - \text{softmin}(d))$, d denotes the distance between each target feature and its corresponding class anchor, and $\circ$ indicates elementwise multiplication. In order to enhance robustness during the early stages of adaptation where the identification of the unknown class is inherently uncertain, in our framework, we introduce a safety margin by applying a conservative scaling factor T to the learned dynamic threshold τ and give the empirical risks of $R_u^s(h)$ as follows:

$$\hat{R}_u^s(h) = \frac{1}{m_s}\sum_{i=1}^{m_s}\begin{cases}\beta_{\min}\left(g\left(\phi\left(x_i^s\right)\right)\right) - T\tau, & \beta_{\min}\left(g\left(\phi\left(x_i^s\right)\right)\right) > T\tau\\ 0, & \text{otherwise}\end{cases} \tag{25}$$

where $\hat{R}_u^t(h)$ follows an identical formulation. This joint mechanism adaptively adjusts the confidence margin for unknown-class rejection according to batch-level uncertainty, leading to more stable threshold calibration across varying domains.

Based on the above analysis, the complete workflow of the proposed JDMUS approach is given in Algorithms 1 and 2.

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Algorithm 2 Cross-Scene Adaptation With JDMUS

Input: Pretrained network (ϕ, g); labeled source dataset D^s; unlabeled target dataset D^t.

Parameter: Regularization parameter λ; kernel bandwidth σ; open-set risk weight γ; joint-distribution weight α; scaling factor T; maximum iteration K_t.

Output: Network $h = g \circ \phi$

for iteration = 1 **to** K_t **do**
  Sample minibatches from D^s and D^t.
  Extract features $\phi_1(x_s)$ and $\phi_1(x_t)$.
  Feed $\phi_1(x_s)$ and $\phi_1(x_t)$ into the CA module.
  Compute distance vectors $\phi(x_s)$ and $\phi(x_t)$.
  Compute target reliability weights w^t.
  Compute $\min(\beta)$ for target features via Eq. (25).
  Compute domain-adversarial outputs.
  Update pseudolabels in D^t using $\min(\beta)$.
  Compute the total loss $\mathcal{L}$ in Eq. (9) on the minibatches.
  Update parameters of ϕ and g via gradient descent.
end for
return adapted model $h = g \circ \phi$.

It consists of source pretraining and cross-scene adaptation in which source pretraining controls source risk (RS), and cross-scene adaptation controls joint distribution discrepancy and open-set difference.

V. Experiments

We implement our algorithm in PyTorch and evaluate it on four cross-scene HSI classification datasets. We compare its performance with state-of-the-art OSDA methods. Sections V.A–V.F introduce the datasets, evaluation metrics, experimental setup, result analysis, parameter sensitivity, ablation study, and statistical test.

A. Datasets

We employ four pairs with the following source–target domain configurations: Pavia University (PU) → Pavia Center(PC), Houston2013 (HU13) → Houston2018 (HU18), Dioni → Loukia, and Ziyuan1-02D Yancheng (ZY) → GaoFen-5 Yancheng (GF). The class definitions and sample counts for each pair are detailed in Tables I–IV of the supplementary material.

1) *PU–PC*: The Pavia datasets are collected by the ROSIS sensor over Pavia, Italy, with a spatial resolution of 1.3 m/pixel. The PU dataset contains 115 spectral bands and covers an area of 610 × 340 pixels, including the university campus and surrounding urban regions. The PC dataset initially includes 115 bands but retains 102 spectral bands after noise removal, with a larger image size of 1096 × 715 pixels, capturing a more complex urban scene. We use the 102 common spectral bands from both datasets for consistency. Seven dominant land-cover classes are selected as the known categories, and one target-domain class (Tiles) is used as the unknown category.
2) *HU13–HU18*: The Houston datasets are collected using the ITRES CASI-1500 airborne hyperspectral sensor. The HU13 dataset, captured in 2013, covers the University of Houston campus and surrounding urban areas, with 144 spectral bands, a spatial resolution of 2.5 m/pixel, and an image size of 349 × 1905 pixels, including 15 land-cover classes. The HU18 dataset, acquired in 2018, features expanded coverage with updated urban landscapes. It contains 48 spectral bands at 1 m/pixel resolution, an image size of 1202 × 4768 pixels, and 20 land-cover classes, enabling high-precision urban land-use studies. For consistency, we use the 48 spectral bands shared by both datasets. Seven representative classes are selected as known categories, with evergreen and deciduous trees in HU18 merged into a single class, while the remaining twelve classes are treated as unknown.
3) *Dioni–Loukia*: The Dioni–Loukia dataset, released by the International Society for Photogrammetry and Remote Sensing (ISPRS), was acquired by the EO-1 satellite. The Dioni subscene has dimensions of 250 × 1376 pixels, while the Loukia subscene measures 249 × 945 pixels. We use the 176 spectral bands common to both datasets. Nine classes from Dioni are selected as known categories, and three classes from Loukia are designated as unknown.
4) *ZY–GF*: The ZY–GF datasets are acquired over China's Yancheng Wetland Reserve, focusing on coastal wetland ecosystem monitoring. The ZY subset has an image size of 1398 × 942 pixels, while the GF subset measures 1175 × 585 pixels. We use the 147 spectral bands common to both datasets. Three representative classes are selected as known categories, and the remaining four classes in GF are treated as unknown.

B. Evaluation Metrics

We use the known-class accuracy overall score (OS*) and unknown-class accuracy (Unk) to evaluate model performance on known and unknown categories, respectively. In addition, the conventional OS is prone to mislead, since it averages accuracy over all classes by treating unknowns as a single class, and underestimates the impact of unknowns (Unk) as the number of known classes increases. To address this, we adopt the Harmonic mean HOS of OS* and Unk, where HOS is defined as

$$\mathrm{HOS} = \frac{2 \times \mathrm{OS}^* \times \mathrm{Unk}}{\mathrm{OS}^* + \mathrm{Unk}}. \tag{26}$$

C. Experimental Setup

All experiments are conducted on a server that runs the Ubuntu 22.04.5 LTS with Linux kernel 5.19.0. The hardware configuration includes dual Intel Xeon E5-2699 v4 processors (22 cores/44 threads per socket, 88 threads in total, base frequency 2.20 GHz, maximum frequency 3.60 GHz), and 503 GB of RAM. The system is also equipped with an NVIDIA GeForce RTX 4080 GPU with 16 GB memory (driver version 535.183.01). The software environment consists of CUDA 11.3, Python 3.7.15, and PyTorch 1.12.1.

The primary hyperparameter configurations of JDMUS are provided as follows.

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TABLE I
UNK (%), OS* (%), AND HOS (%) ON THE PU–PC TASK

Class	Closed-set			Open-set									
	CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	STA	MTS	UADAL	OMEGA	ANNA	WGDT	JDMUS
1	96.4	98.0	94.6	93.2	95.6	93.3	80.0	91.0	64.1	71.3	84.6	**96.1**	92.7
2	99.8	92.1	96.7	84.8	85.7	77.4	67.9	**87.6**	15.3	72.0	65.5	81.0	84.1
3	75.8	80.3	56.7	70.7	**71.7**	68.8	55.0	65.7	39.3	53.7	68.6	47.1	69.9
4	17.8	13.8	25.1	63.1	**77.9**	63.7	54.5	75.7	45.7	25.3	65.0	47.0	63.1
5	100.0	100.0	100.0	98.9	99.2	98.9	96.1	67.3	61.0	91.7	99.0	**100.0**	**100.0**
6	86.2	71.4	88.3	48.0	39.4	29.4	21.2	66.8	55.1	65.8	64.9	64.9	**79.9**
7	83.4	44.1	73.2	44.6	34.3	35.2	38.4	25.8	0.8	18.3	9.2	46.4	**53.9**
UNK (%)	0.0	0.0	0.0	34.3	38.6	34.4	25.8	19.8	**98.6**	64.9	65.9	86.3	87.7
OS* (%)	79.9	71.4	76.4	71.9	72.0	66.7	63.3	68.6	40.2	56.9	65.3	68.9	**77.7**
HOS (%)	0.0±0.0	0.0±0.0	0.0±0.0	46.3±3.7	49.9±6.6	45.3±3.1	35.4±10.6	30.2±7.4	56.1±12.6	59.2±14.4	65.0±4.7	76.5±4.6	**82.4±2.6**

TABLE II
UNK (%), OS* (%), AND HOS (%) ON THE HU13–HU18 TASK

Class	Closed-set			Open-set									
	CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	STA	MTS	UADAL	OMEGA	ANNA	WGDT	JDMUS
1	48.4	94.6	98.4	84.1	74.1	85.9	81.1	88.3	58.6	47.6	**94.2**	74.5	68.7
2	56.5	75.4	42.9	34.6	28.0	31.1	25.7	43.1	33.8	37.0	41.6	34.3	**56.9**
3	64.6	60.1	45.4	62.5	56.2	67.8	**74.7**	73.7	36.8	49.2	51.8	48.1	53.1
4	83.1	99.9	77.3	39.4	61.1	52.1	79.5	80.8	25.6	43.1	56.3	90.6	**99.6**
5	83.9	82.9	77.1	60.5	65.4	65.9	**76.7**	76.5	19.6	35.8	42.3	65.7	66.8
6	47.5	82.7	76.3	33.9	32.2	27.9	27.3	42.9	13.0	35.5	27.2	64.7	**68.9**
7	70.2	35.3	72.8	52.1	51.9	47.4	40.1	**59.4**	8.5	11.3	36.6	3.7	5.9
UNK (%)	0.0	0.0	0.0	26.9	29.6	29.6	24.3	13.2	**89.2**	57.8	56.0	70.2	68.9
OS* (%)	64.9	75.8	70.0	52.4	52.7	54.0	57.7	**66.4**	28.0	37.1	50.0	54.5	59.8
HOS (%)	0.0±0.0	0.0±0.0	0.0±0.0	35.0±5.4	37.6±3.7	37.7±4.9	33.1±8.2	21.4±8.1	41.5±12.5	43.7±9.2	51.7±5.6	61.1±2.8	**64.0±0.4**

1) *Regularization Parameters* λ: Parameters λ are the coefficients of the regularization terms in (15), which help prevent overfitting. We set $\lambda = 1 \times 10^{-4}$ for the PU–PC, HU13–HU18, Dioni–Loukia, and ZY–GF tasks.
2) *Standard Deviation* σ_x: The parameter σ_x in (20) represents the bandwidth of the Gaussian feature kernel function, corresponding to the median of pairwise distances between samples. We choose $\sigma_x = 1$ for the PU–PC, HU13–HU18, Dioni–Loukia, and ZY–GF tasks.
3) *Open-Set Risk Difference Parameter* γ: The hyperparameter γ controls the weight of the term $\hat{R}_u^t(h) - \gamma\hat{R}_u^s(h)$. For this parameter, we set it to 0.01 for PU–PC and HU13–HU18 tasks, and to 0.001 and 0.0001 for Dioni–Loukia and ZY–GF, respectively.
4) *Joint Distribution Weights* α: The hyperparameter α controls the relative importance of JDM between the source and target domains. We set PU–PC and HU13–HU18 to 0.01, and Dioni–Loukia and ZY–GF to 0.001 and 0.0001, respectively.
5) *Scaling Factor* T: The hyperparameter T controls how strictly the open-set empirical risk enforces that unknown-class samples' probabilities deviate from the classification threshold τ. We set $T = 1.1$ for all tasks.

For the remaining hyperparameters, please refer to their default settings in the WGDT framework [18].

D. Results and Analysis

Our method is evaluated against twelve state-of-the-art competitors, including three CSDA baselines: CLDA [9], SCLUDA [10], TAADA [12]; and nine OSDA approaches: OSBP [23], DAMC [36], MTS [38], STA [37], UADAL [17], OMEGA [41], ANNA [40], DACD [39], and WGDT [18].

The quantitative results of cross-scene HSI classification are presented in Tables I–IV. Across four challenging cross-domain classification tasks, the proposed JDMUS framework consistently achieves the highest HOS values, establishing statistically significant improvements over all baselines.

On the PU–PC benchmark (see Table I), JDMUS attains an HOS of 82.4%, surpassing the strongest baseline WGDT (76.5%), by a margin of 5.9%. The per-class analysis indicates that JDMUS achieves the best performance in classes 2, 3, 6, and 7, thereby demonstrating competitiveness on both frequently occurring and challenging categories. Although UADAL achieves the highest unknown-detection rate (UNK = 98.6%), its known-class accuracy is severely compromised (OS* = 40.2%), reflecting an over-specialization toward unknown rejection. By contrast, JDMUS maintains an effective balance, achieving a robust UNK of 87.7% while simultaneously delivering the best OS* of 77.7%, which confirms its ability to jointly optimize known-class classification and unknown detection.

The HU13–HU18 benchmark (see Table II) further underscores the balanced generalization capability of JDMUS. It achieves an HOS of 64.0%, exceeding WGDT (61.1%) by 2.9%. The baselines exhibit polarized behavior: MTS attains the highest OS* (66.4%) but fails on unknown detection (UNK = 13.2%), while UADAL achieves the strongest UNK (89.2%) at the expense of known-class accuracy (OS* = 28.0%).

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TABLE III
UNK (%), OS* (%), AND HOS (%) ON THE DIONI–LOUKIA TASK

Class	Closed-set			Open-set									
	CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	STA	MTS	UADAL	OMEGA	ANNA	WGDT	JDMUS
1	0.7	4.4	48.4	12.9	**24.8**	19.3	9.5	16.9	4.9	10.8	8.5	3.9	1.5
2	100.0	71.1	95.4	48.1	43.5	57.2	68.7	83.9	96.7	**97.2**	87.6	88.0	92.5
3	59.4	26.2	72.5	21.7	33.4	31.3	41.5	28.5	28.0	19.3	21.7	51.5	**83.8**
4	93.0	100.0	79.9	70.4	72.0	70.4	76.5	**84.1**	55.7	39.4	49.4	56.3	67.1
5	0.0	0.1	36.6	8.0	13.1	13.2	**22.3**	7.9	2.5	17.6	2.9	0.1	0.0
6	81.3	64.0	56.2	62.8	58.9	56.9	39.8	**73.3**	36.4	38.1	57.7	63.7	72.7
7	18.3	16.3	50.0	11.6	**17.1**	11.7	3.5	11.1	3.0	2.8	11.8	5.7	5.4
8	6.3	72.1	45.6	3.1	2.0	4.7	7.4	24.7	38.1	10.9	10.3	52.7	**58.9**
9	100.0	97.4	99.8	28.5	8.7	18.0	61.8	82.3	85.7	83.6	14.8	97.7	**100.0**
UNK (%)	0.0	0.0	0.0	23.4	25.0	23.0	13.9	6.2	48.8	**54.0**	46.4	52.0	48.3
OS* (%)	51.0	50.2	64.9	29.7	30.4	31.4	36.8	45.6	39.0	35.5	29.4	46.6	**53.6**
HOS (%)	0.0±0.0	0.0±0.0	0.0±0.0	25.3±5.2	26.4±3.1	25.7±5.3	18.5±10.2	10.3±7.0	41.6±4.3	40.4±10.3	34.0±6.1	48.9±1.3	**50.8±0.6**

TABLE IV
UNK (%), OS* (%), AND HOS (%) ON THE ZY–GF TASK

Class	Closed-set			Open-set									
	CLDA	SCLUDA	TAADA	DACD	OSBP	DAMC	STA	MTS	UADAL	OMEGA	ANNA	WGDT	JDMUS
1	100.0	100.0	100.0	85.8	89.2	82.6	72.9	90.6	62.3	67.8	29.1	83.7	**95.6**
2	59.2	100.0	100.0	**100.0**	**100.0**	**100.0**	91.6	80.8	15.2	65.4	53.7	99.6	**100.0**
3	97.6	100.0	82.4	93.3	91.6	**99.5**	92.5	75.0	80.7	77.8	14.8	81.2	65.4
UNK (%)	0.0	0.0	0.0	13.4	15.5	14.5	49.2	51.0	58.8	48.3	76.4	90.2	**94.5**
OS* (%)	85.6	100.0	94.1	93.0	93.6	**94.0**	85.7	82.1	52.7	70.4	32.5	88.2	90.3
HOS (%)	0.0±0.0	0.0±0.0	0.0±0.0	23.4±4.3	26.6±0.7	25.0±1.1	60.4±12.1	62.7±11.9	49.3±9.8	55.7±9.3	43.6±8.2	88.9±3.8	**92.3±1.4**

JDMUS bridges this gap by achieving a favorable trade-off (OS* = 59.8% and UNK = 68.9%), resulting in the most reliable overall performance. It is noteworthy that JDMUS provides leading accuracy in classes 2, 4, and 6, highlighting its robustness across heterogeneous land-cover types.

The Dioni–Loukia task (see Table III) represents a particularly challenging scenario with a larger label space and substantial domain discrepancy. JDMUS achieves the best HOS of 50.8%, outperforming all baselines. Although TAADA provides the highest OS* (64.9%) and OMEGA excels in unknown detection (UNK = 54.0%), both methods suffer from unbalanced performance. JDMUS achieves the best results in classes 3, 8, and 9, while delivering competitive outcomes in classes 2, 4, and 6, thereby confirming its ability to handle imbalanced distributions and complex interclass variability.

On the most demanding ZY–GF task (see Table IV), JDMUS demonstrates remarkable robustness, achieving an HOS of 92.3%, which significantly outperforms WGDT (88.9%). A closer inspection reveals that SCLUDA attains a perfect OS* of 100.0% but fails entirely in unknown detection (UNK = 0.0%), rendering it unsuitable for practical HSI OSDA applications. In sharp contrast, JDMUS delivers the highest UNK (94.5%) while simultaneously achieving an OS* of 90.3%, thereby ensuring reliable performance in real-world open-set environments. JDMUS achieves top-ranked accuracy in classes 1 and 2, providing further evidence of its superior discriminative capability.

The visual classification results are illustrated in Figs. 1–4 of the supplementary material, which show that JDMUS produces cleaner and more coherent land-cover predictions than competing methods. Taken together, these results establish that JDMUS consistently delivers balanced improvements across all evaluation tasks and metrics, effectively addressing the long-standing trade-off between known-class accuracy and unknown-class detection. This robustness under diverse and challenging scenarios validates JDMUS as a reliable framework for cross-domain open-set HSI classification.

E. Parameter Sensitivity and Ablation Study

In this part, extensive experiments are conducted on four distinct tasks to systematically examine the influence of parameters (λ, α, γ, T, and σ_x) on JDMUS performance. Via grid search, the parameter combinations are within the following ranges: $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, $\alpha \in \{0.1, 0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001, 0.00005, 0.00001\}$, and $\gamma \in \{0.1, 0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001, 0.00005, 0.00001\}$. T serves as a safety margin that controls the separation between known and unknown classes within the dynamic thresholding process. Theoretically, T can be derived from the bound on the false-accept rate of unknown samples under a mild concentration assumption on prediction confidence. Empirically, we tune T within a small range $\{1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7\}$ on the validation split of the source domain. Meanwhile, the kernel bandwidth σ_x, corresponding to the Gaussian feature kernel in (20), is determined via the median-distance heuristic and normalized by the feature dimension to ensure scale invariance.

From Fig. 3(a), we can notice that while holding other parameters constant, the optimal α values differ across tasks: 0.001 for both PU–PC and Dioni–Loukia, 0.0001 for HU13–HU18, and 0.00001 for ZY–GF. Fig. 3(b) illustrates

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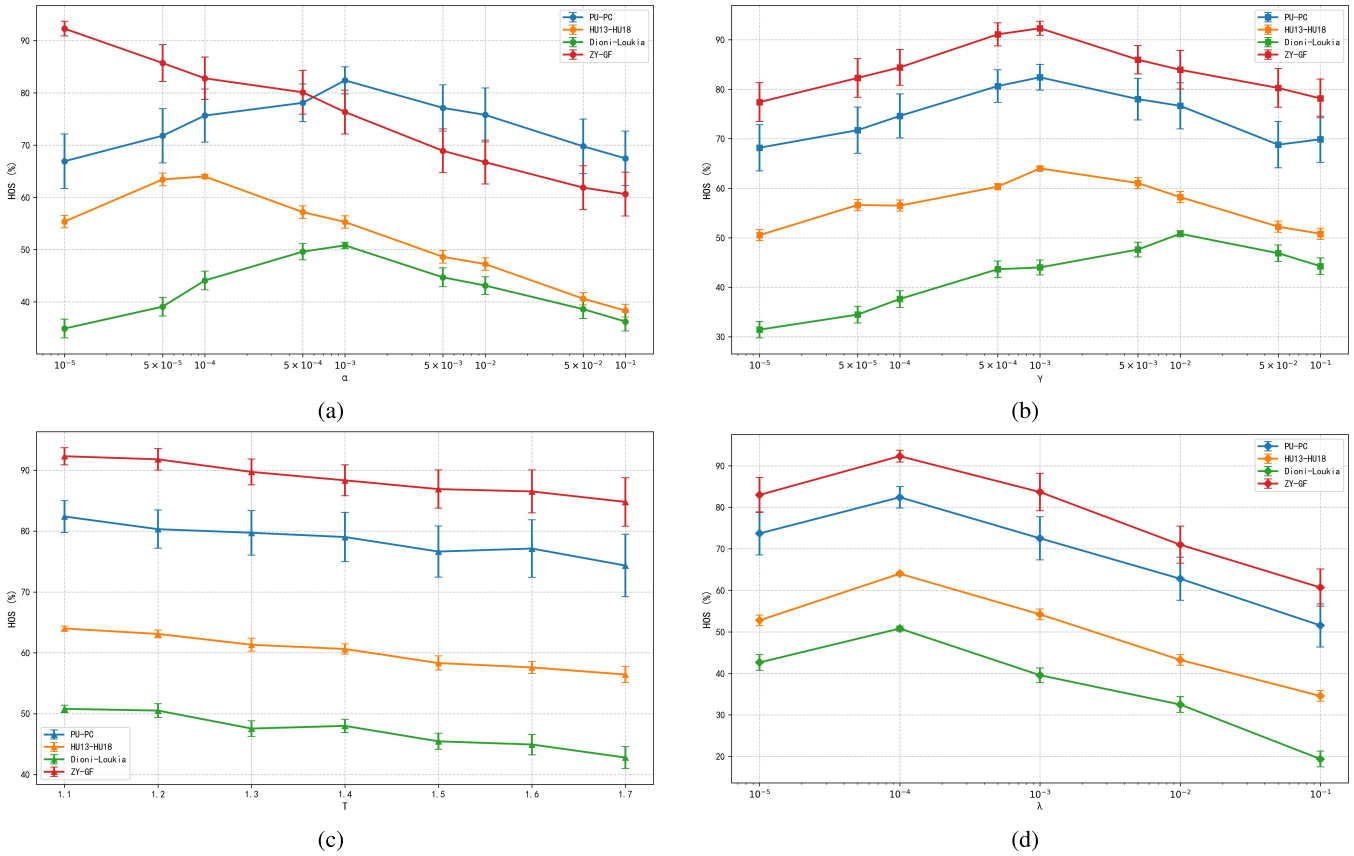


Fig. 3. Parameter sensitivity analysis. (a) HOS versus α, (b) HOS versus γ, (c) HOS versus T, and (d) HOS versus λ.

TABLE V
ABLATION STUDY OF JDMUS

Method	PU-PC	HU13-HU18	Dioni-Loukia	ZY-GF
RS + US	63.0±5.9	55.6±5.3	43.1±3.3	72.5±11.2
RS + JDA	79.3±3.5	60.2±2.9	47.3±1.7	87.1±3.5
JDMUS	**82.4±2.6**	**64.0±0.4**	**50.8±0.6**	**92.3±1.4**

that the highest HOS scores occur at $\gamma = 0.001$ for PU–PC, HU13–HU18, and ZY–GF, whereas Dioni–Loukia achieves peak performance at $\gamma = 0.01$. Furthermore, Fig. 3(c) demonstrates consistent optimal performance across all tasks when $T = 1.1$, yielding maximum HOS values in each task. In addition, as shown in Fig. 3(d), the optimal value of the ridge regularization coefficient λ is consistently 10^{-4} across all four cross-scene tasks.

Meanwhile, as illustrated in Table V, we conduct ablation experiments to evaluate the individual contributions of the RS, JDM, and US modules within the JDMUS framework. Three ablated configurations are constructed: JDMUS without JDM (RS+US), JDMUS without US (RS+JDA), and the full JDMUS model. All methods run independently ten times under each configuration. Table V reports the mean and standard deviation of OA scores across the PU–PC, HU13–HU18, Dioni–Loukia, and ZY–GF tasks. The complete JDMUS consistently achieves the highest OA on all tasks, underscoring the necessity of each module.

Notably, the absence of the JDM module leads to a considerable performance drop, affirming its critical role in aligning feature distributions across domains and facilitating knowledge transfer. Similarly, removing the US module also results in degraded accuracy, with OA reductions of 3.1%, 3.8%, 3.5%, and 5.2% across the four tasks, illustrating its utility in enhancing feature discrimination and robustness. While the gain from the US is comparatively smaller than that from JDM, its consistent positive effect across datasets confirms its supplementary value. These results collectively demonstrate that the synergistic combination of all three modules optimally supports cross-scene classification for remote sensing imagery.

F. Statistical Test

To ensure statistical soundness, we comprehensively evaluate whether the proposed JDMUS method achieves significantly better performance than all open-set baseline algorithms. We focus on the ten open-set approaches, excluding the three closed-set methods (CLDA, SCLUDA, and TAADA) from statistical testing.

The average ranks of the algorithms are presented in Table VI. Based on these rankings, the Friedman statistic is computed as

$$\chi_F^2 = \frac{12N}{k(k+1)}\left[\sum_{i=1}^{k} R_i^2 - \frac{k(k+1)^2}{4}\right]$$

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TABLE VI
HOLM'S POST HOC ANALYSIS: JDMUS VERSUS OPEN-SET BASELINES

i	**Method**	**Rank** R_i	z	p	$\alpha/(k-i)$	**Rej.**
1	JDMUS	1.00	–	–	–	–
2	WGDT	2.00	0.47	0.320	0.0056	No
3	OMEGA	4.25	1.52	0.064	0.0063	Yes
4	ANNA	4.50	1.64	0.051	0.0071	Yes
5	UADAL	4.75	1.75	0.040	0.0083	Yes
6	OSBP	7.50	3.04	0.001	0.0100	Yes
7	DAMC	8.00	3.27	0.001	0.0125	Yes
8	STA	8.25	3.39	0.001	0.0167	Yes
9	DACD	9.50	3.97	<0.001	0.0250	Yes
10	MTS	9.75	4.09	<0.001	0.0500	Yes

$$F_F = \frac{(N-1)\chi_F^2}{N(k-1)-\chi_F^2} \tag{27}$$

where $N = 12$ is the number of tasks and $k = 10$ is the number of open-set algorithms. $\chi_F^2 = 41.88$, and $F_F = 6.967$. The F_F value significantly exceeds the critical value $F(9, 9 \times 11) = 1.97$ at a significant level $\alpha = 0.05$. Therefore, the null hypothesis of equal performance among all open-set algorithms is statistically rejected.

To determine whether our proposed algorithm can significantly perfrom better than the other baselines, we conduct Holm's step-down post hoc analysis. The pairwise hypotheses are ordered by their z-statistics. For each comparison, the statistic is computed as

$$z_i = \frac{R_i - R_{\text{JDMUS}}}{\sqrt{k(k+1)/(6N)}} \tag{28}$$

with corresponding p-values obtained from the standard normal distribution. Each hypothesis is then evaluated against its Holm-adjusted critical value $\alpha/(k-i)$.

As demonstrated in Table VI, JDMUS demonstrates statistically significant improvements over eight of the nine baseline methods. To better compare our proposed JDMUS and the baseline WGDT, we conduct t-test for them. The null hypothesis assumes no difference in the mean performance. The statistic p-value is equal to 0.009, and below 0.05. Therefore, the null hypothesis is rejected. The mean value of JDMUS is 72.53, while that of WGDT is 69.35. Based on the above analysis, the proposed JDMUS is statistically superior to all comparison algorithms in open-set HSI classification.

VI. CONCLUSION

In this article, we propose a novel OSDA method, JDMUS, for cross-scene open-set HSI classification. In JDMUS, we theoretically derive a tighter generalization error bound based on joint probability distribution matching and provide its exact closed-form solution. Under this new error bound, JDMUS establishes a JDM mechanism to enable reliable knowledge transfer at both the semantic and feature levels, and designs an unknown-class US mechanism to avoid the classification performance fluctuation caused by blindly recognizing target data as unknown classes, thus exhibiting high interpretability and generalization capability. Extensive experiments demonstrate that compared to the SOTA deep HSI OSDA methods, JDMUS can better identify unknown samples while maintaining excellent HSI classification accuracy on known classes.

In real-world remote sensing applications, data are often collected from heterogeneous sources, such as different satellites, sensors, or geographic regions, and inevitably contain both known and unknown categories. In the future, we will extend the JDMUS framework proposed in this work to the more challenging setting of multisource OSDA (MSOSDA). This extension will allow for an in-depth investigation into knowledge transfer and unknown-class recognition within multisource data, thereby more fully leveraging the application value of massive remote sensing data.

APPENDIX
PROOF OF THEOREM 1: GENERALIZATION ERROR BOUNDS FOR OSDA

A. Preliminaries and Definitions

The risks for known classes and unknown classes in the target domain are defined as $R^{tk}(h)$ and $R^{tu}(h)$, respectively,

$$R^{tk}(h) = \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y) P_{X^tY^t}(x, y)\, dxdy \tag{29}$$

$$R^{tu}(h) = \int_{\mathcal{X}\times\mathcal{Y}^u} \ell(h(x), y) P_{X^tY^t}(x, y)\, dxdy. \tag{30}$$

To establish the generalization error bound in Theorem 1, we employ a *divide-and-conquer* strategy by decomposing the target domain risk $R^t(h)$ into two complementary components

$$R^t(h) = \underbrace{R^{tk}(h)}_{\text{Known-Classes Risk}} + \underbrace{R^{tu}(h)}_{\text{Unknown-Classes Risk}}. \tag{31}$$

The proof proceeds in three logical steps as follows.

1) Bound the known-classes risk $R^{tk}(h)$ via distribution alignment (see Theorem 2).
2) Bound the unknown-classes risk $R^{tu}(h)$ via open-set identification analysis (see Theorem 3).
3) Combine both bounds to obtain the final generalization error bound.

B. Step 1: Bounding Known Classes Risk via Distribution Alignment

Theorem 2 (Approximate Alignment of Joint Distributions for Known Classes): The following inequality holds:

$$R^{tk}(h) \le R^s(h) + M\sqrt{2D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \,\|\, P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)} \tag{32}$$

where $P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}$ denotes the joint distribution of the target domain, restricted to the subspace of known classes $\mathcal{X}\times\mathcal{Y}^s$.

Proof: We begin by quantifying the discrepancy between the source and target known class distributions. Define the total variation (TV) distance and Chi-Square (χ^2) divergence between two distributions P and Q

$$D_{TV}(P \,\|\, Q) = \int |P(x, y) - Q(x, y)|\, dxdy \tag{33}$$

$$D_{CS}(P \,\|\, Q) = \int P(x, y)\frac{P(x, y)}{Q(x, y)} dxdy - 1. \tag{34}$$

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Assumption 1: The loss function $\ell(h(x), y) \leq M$ for all $y \in \mathcal{Y}^s$, where $M > 0$

$$\begin{aligned}
&\left|R^{tk}(h) - R^s(h)\right| \\
&= \left| \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y) \left[P_{X^tY^t}(x,y) - P_{X^sY^s}(x,y)\right] dxdy \right| \\
&\leq \int_{\mathcal{X}\times\mathcal{Y}^s} \left| \ell(h(x), y) \left[P_{X^tY^t}(x,y) - P_{X^sY^s}(x,y)\right] \right| dxdy \\
&\leq M \int_{\mathcal{X}\times\mathcal{Y}^s} \left|P_{X^tY^t}(x,y) - P_{X^sY^s}(x,y)\right| dxdy \qquad (35) \\
&= M D_{TV}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right) \qquad (36) \\
&\leq M\sqrt{2D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)}. \qquad (37)
\end{aligned}$$

From Assumption 1, we obtain (35); from the definition of TV distance (33), we obtain (36); and from the f-divergence inequality $D_{TV}(P \parallel Q) \leq (2D_{\chi^2}(P \parallel Q))^{1/2}$ we obtain (37). Therefore,

$$\begin{aligned}
R^{tk}(h) \leq\ & R^s(h) \\
&+ M\sqrt{2D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)}.
\end{aligned}$$

C. Step 2: Bounding Unknown-Classes Risk via Open-Set Analysis

Theorem 3: (Precise Identification of Target Domain Unknown-Class Samples): The following equation holds:

$$\begin{aligned}
R^{tu}(h) \leq\ & R^t_u(h) - R^s_u(h) \\
&+ N\sqrt{2D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)}. \qquad (38)
\end{aligned}$$

Proof: We analyze the open-set risk functions, which measure the tendency to classify samples as "unknown."

Assumption 2: The loss function $\ell(h(x), y^u) \leq N$ for $y^u \in \mathcal{Y}^u$, where $N > 0$.

From (5), the risk of target samples being labeled as unknown is

$$\begin{aligned}
R^t_u(h) &= \int_{\mathcal{X}} \ell(h(x), y^u) P_{X^t}(x)\, dx \\
&= \int_{\mathcal{X}\times\mathcal{Y}^u} \ell(h(x), y^u) P_{X^tY^t}(x,y)\, dxdy \\
&\quad + \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^tY^t}(x,y)\, dxdy. \qquad (39)
\end{aligned}$$

From (4), the risk of source samples being labeled as unknown is

$$\begin{aligned}
R^s_u(h) &= \int_{\mathcal{X}} \ell(h(x), y^u) P_{X^s}(x)\, dx \\
&= \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^sY^s}(x,y)\, dxdy. \qquad (40)
\end{aligned}$$

Subtracting (39) from (40),

$$\begin{aligned}
&R^t_u(h) - R^s_u(h) \\
&= \int_{\mathcal{X}\times\mathcal{Y}^u} \ell(h(x), y^u) P_{X^tY^t}(x,y)\, dxdy \\
&\quad + \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^tY^t}(x,y)\, dxdy \\
&\quad - \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^sY^s}(x,y)\, dxdy \\
&= R^{tu}(h) + \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^tY^t}(x,y)\, dxdy \\
&\quad - \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^sY^s}(x,y)\, dxdy. \qquad (41)
\end{aligned}$$

Using Assumption 2 and following a similar bounding technique as in Theorem 1,

$$\begin{aligned}
&\left| \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^tY^t}(x,y)\, dxdy \right. \\
&\quad \left. - \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^sY^s}(x,y)\, dxdy \right| \\
&\leq \int_{\mathcal{X}\times\mathcal{Y}^s} \left| \ell(h(x), y^u) \left[P_{X^tY^t}(x,y) - P_{X^sY^s}(x,y)\right] \right| dxdy \\
&\leq N \int_{\mathcal{X}\times\mathcal{Y}^s} \left|P_{X^tY^t}(x,y) - P_{X^sY^s}(x,y)\right| dxdy. \qquad (42)
\end{aligned}$$

From (41) and (42), we have:

$$\begin{aligned}
R^{tu}(h) &= R^t_u(h) - R^s_u(h) \\
&\quad - \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^tY^t}(x,y)\, dxdy \\
&\quad + \int_{\mathcal{X}\times\mathcal{Y}^s} \ell(h(x), y^u) P_{X^sY^s}(x,y)\, dxdy \\
&\leq R^t_u(h) - R^s_u(h) \\
&\quad + N\sqrt{2D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)}. \qquad (43)
\end{aligned}$$

D. Step 3: Final Combination for Theorem 1

Proof of Theorem 1: Combining the results from Theorems 2 and 3, we now substitute the established bounds into the target risk decomposition given by (31)

$$\begin{aligned}
R^t(h) &= R^{tk}(h) + R^{tu}(h) \\
&\leq R^s(h) + M\sqrt{2D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)} \\
&\quad + R^t_u(h) - R^s_u(h) \\
&\quad + N\sqrt{2D_{CS}\left(P_{X^sY^s\sim s\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)} \\
&= R^s(h) + R^t_u(h) - R^s_u(h) \\
&\quad + (M+N)\sqrt{2D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)} \qquad (44)
\end{aligned}$$

thus, under Assumptions 1 and 2, for any hypothesis $h : \{\mathcal{X} \to \mathbb{R}^{|\mathcal{Y}^t|} \to \mathbb{R}^{C_s+1}\}$, the generalization error on the target domain satisfies the following bound:

$$\begin{aligned}
R^t(h) \leq\ & R^s(h) + R^t_u(h) - R^s_u(h) \\
&+ (M+N)\sqrt{2D_{CS}\left(P_{X^sY^s\sim\mathcal{X}\times\mathcal{Y}^s} \parallel P_{X^tY^t\sim\mathcal{X}\times\mathcal{Y}^s}\right)}. \qquad (45)
\end{aligned}$$

This completes the proof of Theorem 1.

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